TARGETED DRILL | ANALYTICS-DA-201

Calculations, LODs and Functions

20-question targeted drill

Questions	20 single-answer multiple-choice
Recommended time	35 minutes (about 1.75 minutes per question)
Focus	LOD grain, filter order, aggregate vs. row-level, functions
Product version	Tableau 2024.2
Format	Closed book | no Tableau | no notes

The GATE method

G	**Grain**	What rows must the number see? Row, customer, region, company?
A	**Aggregation**	Row-level, aggregate, LOD, or table calculation?
T	**Timing**	Before or after filters? Must hidden rows count?
E	**Example**	Test with 2-3 rows. Does the result make sense?

Before each question: write G and T in the margin. For a ratio, write the grain of the numerator and the grain of the denominator separately. Do not look at the options first.

Independent study material aligned to the published Analytics-DA-201 objectives. Not an official Salesforce or Tableau exam. Behavior reflects Tableau 2024.2.

Answer Sheet

Write the letter of your answer for each question.

Q	Q	Q	Q
1 _____	**6** _____	**11** _____	**16** _____
2 _____	**7** _____	**12** _____	**17** _____
3 _____	**8** _____	**13** _____	**18** _____
4 _____	**9** _____	**14** _____	**19** _____
5 _____	**10** _____	**15** _____	**20** _____

Scoring

One point per question. 18+ / 20 is strong. For every miss, write whether the error was **grain** (wrong rows), **timing** (wrong filter stage), or **function** (wrong syntax). Grain and timing errors matter more than syntax.

Questions 1-20

LODs, calculations, functions, and formulas | GATE first, formula second

QUESTION 1 **SINGLE ANSWER**

A view shows **Category** and `SUM([Sales])`, with a normal dimension filter `Region = East`. The calculation is:

```
SUM([Sales]) / SUM({FIXED : SUM([Sales])})
```

What does the denominator represent?

A. Total sales for the East region only.

B. Total sales for the selected category.

C. Total sales for all regions in the data source.

D. Total sales for the categories currently visible.

QUESTION 2 **SINGLE ANSWER**

A view contains **Region** and **Category**. The analyst needs each category's sales as a share of its **region's total**.

Which calculation is correct?

A. `SUM([Sales]) / SUM({FIXED [Category] : SUM([Sales])})`

B. `SUM([Sales]) / SUM({EXCLUDE [Category] : SUM([Sales])})`

C. `SUM([Sales]) / SUM({INCLUDE [Category] : SUM([Sales])})`

D. `SUM([Sales]) / SUM({FIXED : SUM([Sales])})`

QUESTION 3 **SINGLE ANSWER**

A view shows **Region** and `SUM([Sales])`. The analyst needs the **average total sales per customer** for each region.

Which calculation is correct?

A. `AVG({INCLUDE [Customer ID] : SUM([Sales])})`

B. `SUM([Sales]) / COUNT([Customer ID])`

C. `AVG([Sales])`

D. `{EXCLUDE [Customer ID] : AVG([Sales])}`

QUESTION 4 **SINGLE ANSWER**

What does this calculation return?

```
{FIXED [Region] : MAX({FIXED [Region], [City] : SUM([Sales])})}
```

A. Total sales for each region.

B. The average city-level sales across all regions.

C. The highest city-level sales within each region.

D. The sum of each city's maximum sales.

QUESTION 5 **SINGLE ANSWER**

A region has **three category rows**, each with `SUM(Sales) = 300`. The region's true total is 900. The view shows only **Region**, and the field is:

```
M = SUM({FIXED [Region] : SUM([Sales])})
```

What does the region's mark display?

A. 900

B. 1,800

C. 2,700

D. 300

QUESTION 6 **SINGLE ANSWER**

A view shows **Region** and `{FIXED [Region] : SUM([Sales])}`. The analyst adds a **measure filter** `SUM([Sales]) > 1,000`, which hides one region's mark.

What happens to the FIXED LOD values for the regions that remain visible?

A. They are recalculated using only the rows that pass the measure filter.

B. They become NULL for every region.

C. The measure filter is ignored completely.

D. They remain the full region totals, because FIXED is computed before measure filters.

QUESTION 7 **SINGLE ANSWER**

A view shows **Region** and **Category** with `{FIXED [Region] : SUM([Sales])}`. The analyst filters `Category = Office` and adds that filter to **context**.

What happens to each region's FIXED value?

A. It stays the same because FIXED ignores all filters.

B. It becomes NULL because context filters break FIXED LODs.

C. It is recalculated using only Office-category rows.

D. It becomes the grand total across all categories.

QUESTION 8 **SINGLE ANSWER**

Some rows have a NULL `Region`. The calculation `{FIXED [Region] : SUM([Sales])}` is used in a view.

How does Tableau treat the NULL region?

A. NULL region rows are dropped from the LOD calculation.

B. NULL rows are folded into the grand total for every region.

C. The calculation returns an error until NULLs are removed.

D. NULL is treated as its own partition and gets its own LOD value.

QUESTION 9 **SINGLE ANSWER**

Three order lines contain:

- Sales 100, Quantity 2
- Sales 300, Quantity 4
- Sales 200, Quantity 1

Management wants the **weighted revenue per unit across all orders**. Which expression returns that value?

A. `AVG([Sales] / [Quantity])`

B. `SUM([Sales] / [Quantity])`

C. `SUM([Sales]) / SUM([Quantity])`

D. `AVG([Sales]) * AVG([Quantity])`

QUESTION 10 **SINGLE ANSWER**

A table has **5 order lines**, **3 distinct Order IDs**, **2 distinct customers**, and total Sales = 600.

Which expression returns the **average sales per distinct order**?

A. `SUM([Sales]) / COUNTD([Order ID])`

B. `SUM([Sales]) / COUNT([Sales])`

C. `AVG([Sales])`

D. `SUM([Sales]) / COUNTD([Customer ID])`

QUESTION 11 **SINGLE ANSWER**

A view shows one mark per **Category**. Each category contains several different **Region** values. The analyst adds `ATTR([Region])`.

What does the mark display for Region?

A. The first Region alphabetically.

B. An asterisk (*).

C. NULL.

D. A comma-separated list of all Regions.

QUESTION 12 **SINGLE ANSWER**

What does this calculation return?

```
DATEDIFF('month', MAKEDATE(2025, 1, 31), MAKEDATE(2025, 3, 1))
```

A. 2

B. 1

C. 3

D. 0

QUESTION 13 **SINGLE ANSWER**

Which calculation returns the **last day of the previous month** relative to today?

A. `DATETRUNC('month', DATEADD('day', -1, TODAY()))`

B. `DATEADD('day', -1, DATETRUNC('month', TODAY()))`

C. `DATEADD('month', -1, TODAY())`

D. `DATETRUNC('month', TODAY())`

QUESTION 14 **SINGLE ANSWER**

The field `[ID]` contains `INV-2025-0042-X`. Which calculation returns exactly `0042`?

A. `LEFT([ID], 4)`

B. `MID([ID], 10, 4)`

C. `RIGHT([ID], 6)`

D. `FIND([ID], '-')`

QUESTION 15 **SINGLE ANSWER**

What does `REPLACE('banana', 'a', 'o')` return?

A. bonana

B. banano

C. banana

D. bonono

QUESTION 16 **SINGLE ANSWER**

`[Ship Date]` is NULL for some rows. What does this return for those rows?

```
IIF([Ship Date] > TODAY(), 'Late', 'On Time', 'Unknown')
```

A. `Late`

B. `Unknown`

C. `On Time`

D. NULL

QUESTION 17 **SINGLE ANSWER**

`[Country]` is a text field containing some NULL values. Which calculation replaces NULL with the text `Unknown`?

A. `ZN([Country])`

B. `ISNULL([Country])`

C. `IF [Country] = NULL THEN 'Unknown' END`

D. `IFNULL([Country], 'Unknown')`

QUESTION 18 **SINGLE ANSWER**

Which calculation returns `-8`?

A. `FLOOR(-7.2)`

B. `INT(-7.2)`

C. `ROUND(-7.2, 0)`

D. `CEILING(-7.2)`

QUESTION 19 **SINGLE ANSWER**

A view has **Year** and **Month** on Columns and `SUM(Sales)` on Rows. The analyst needs a running total that **restarts each year**.

Which table calculation configuration is correct?

A. Address by Year and partition by Month.

B. Compute Using = Table (across) with no partition.

C. Use `SUM([Sales])` and add a Year quick filter.

D. Address by Month and partition by Year.

QUESTION 20 **SINGLE ANSWER**

An analyst wants to create a **10-mile delivery-radius polygon** around each store using the store's latitude and longitude.

Which calculation should be used?

A. `BUFFER(MAKEPOINT([Lat], [Lon]), 10, 'mi')`

B. `MAKEPOINT([Lat], [Lon])`

C. `MAKELINE(MAKEPOINT([Lat], [Lon]), MAKEPOINT([Lat], [Lon]))`

D. `DISTANCE(MAKEPOINT([Lat], [Lon]), MAKEPOINT([Lat], [Lon]), 'mi')`